

riteangle — Men’s AI Photo Engine

Vendor & Approach Evaluation · 29 Jun 2026

DECISION

Ship **Google Gemini 2.5 Flash Image** with **multi-reference + edit-framing + a flattering dial** as the MVP photo engine. **No per-person training**. ~\$0.04/image → ~\$120 per 1,000 men (3 photos each). Per-person fine-tuning (Astria) is reserved as an optional paid “Pro photos” tier.

Objective & constraints

riteangle never shows a man’s raw photo; viewers only ever see an AI-enhanced portrait built from his own uploads. We needed an engine that is, in priority order: (1) **unmistakably him**, (2) **flattering** (a dating photo, his genuine best), (3) **photorealistic**, and (4) affordable at MVP scale — the target was ~\$100 for 1,000 men. We tested six approaches across four vendors on a cohort of five diverse real men.

Approaches tested — at a glance

Approach	Identity	Realism	Cost / image	Verdict
Gemini 2.5 — edit-framing + flattering	Strong	High	~\$0.04	SELECTED ★
Gemini 2.5 — “recreate” framing	Poor — drifts/ages/baldens	High	~\$0.04	Rejected
Astria — per-person fine-tune (FLUX LoRA)	Best	High	\$1.50 train +\$0.10	Premium tier only
flux-pulid (FLUX, Replicate)	Medium	Medium — glossy	~\$0.04	Rejected — baldens, glossy
InstantID (SDXL, Replicate)	Medium (needs hair prompt)	Poor — cartoon, bad hands	~\$0.08	Rejected
fal.ai flux-pulid (previous prod engine)	—	—	~\$0.05	Account out of balance
Grok / xAI	—	—	—	Not tested (unfunded)

Identity scoring is human-judged (the subject’s own face is the ground truth); automated vision scores under-rate correct-but-different-angle shots.

Same man, six approaches

All generated from the same reference photo of “Chris” (long curly hair, full beard) — the hardest identity case. Compare each against the real reference at top-left.



REAL reference



Gemini “recreate” — drifts, de-ages



InstantID — generic short hair



InstantID + hair prompt —
cartoonish



flux-pulid — balds, glossy



Astria fine-tune — best, but \$\$\$



Gemini edit-framing — SELECTED

Per-approach result

- **Gemini 2.5 — “recreate” framing.** Most realistic of the field, but prompting it to “recreate him” made it idealize — drifting to an older, balder, or different face. **Rejected.**
- **InstantID (SDXL).** Locks the face but its embedding ignores hair, so it produced generic short hair; adding hair to the prompt restored the curls but the output was cartoonish with broken hands and over-muscling. **Rejected.**
- **flux-pulid (FLUX).** Photorealistic and good build, but receded/balded his hairline and looked glossy; identity only approximate. **Rejected.**
- **Astria — per-person fine-tune.** The only zero-effort path to a perfect result: realistic and exactly him, full hair. But ~\$1.50/man to train + a 10–30 min wait. **Held back as a paid “Pro” tier.**
- **Gemini 2.5 — edit-framing + flattering.** Framing it as an **edit of one real man** (“keep him identical, change only the scene”) preserved identity + hair while keeping realism; a flattering clause makes it his most attractive self. Realistic, his hair, looks like him, ~\$0.04/img. **Selected.**

Cohort validation — the winning recipe across 5 diverse men

To prove the recipe is not tuned to one face, we ran it on five very different men. Each pair below is the man's real reference (left) and his AI result (right).



reference



AI result

man01 — long curly hair, beard
Holds — “the very best”



reference

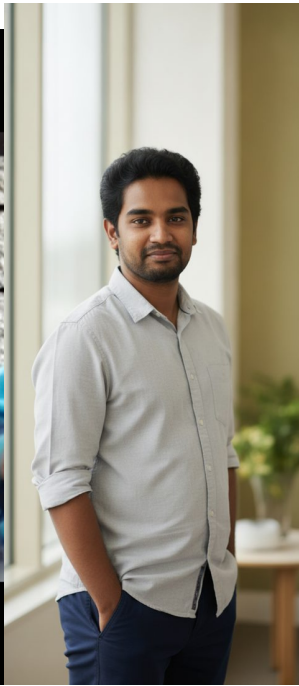


AI result

man02 — young, clean-shaven
Holds



reference



AI result

man03 — short hair, poor source photos
Holds (robust to bad inputs)

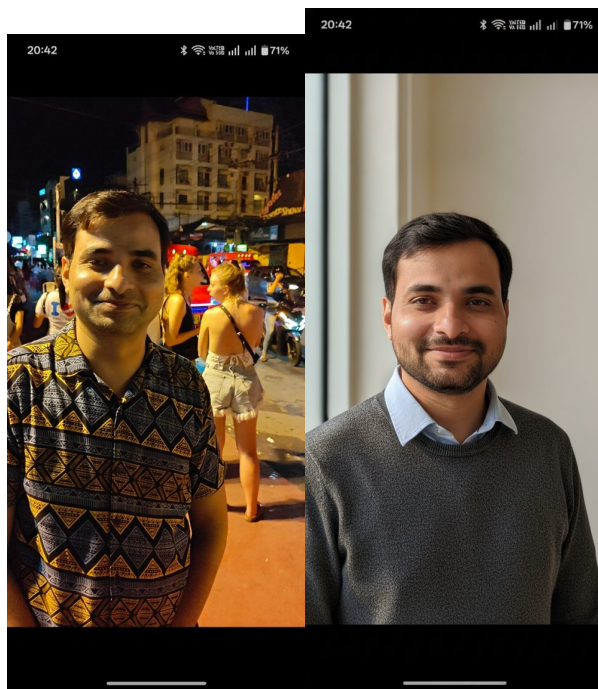


reference



AI result

man04 — older, greying, glasses
Edge case — needs grey-hair guard



reference

AI result

man05 — middle-aged, dark hair
Holds

Result: strong on 4 of 5; one fixable edge case (older grey-haired men de-age unless explicitly guarded). The recipe generalizes across age, hair type, and source-photo quality.

Key findings & production guardrails

Every guardrail below came from a real failure during testing:

- **Prompt as an EDIT, not a recreation.** The single biggest lever. “Recreate him” drifts; “keep this exact man, change only the scene” preserves identity.
- **Flattering dial.** Pure fidelity kept eye-bags, dark circles, grey stubble — realistic but unattractive. Add a “show him at his best” clause (soften under-eye, groom, good light) without reshaping or de-aging.
- **Reference-photo selection is make-or-break.** Feeding all of a man’s photos (incl. wet-hair / hats / group shots) balded him; a curated set of clean, frontal, dry-hair, solo photos produced “the very best.” Auto-select good refs; reject the rest.
- **Styling by body feature.** Genuinely bald → add a stylish hat; heavier build / pot belly → a structured blazer. Driven off a per-man vision read. Never hat a man with good hair.
- **Waist-up framing.** Full-body shots risk an oversized head / wrong proportions. Constrain to head-and-shoulders → waist-up.
- **Older / grey men need an explicit guard.** “Preserve his grey hair and real age” — otherwise the model darkens hair and de-ages.
- **Generate-N, pick-best.** Output varies shot-to-shot; generate several per scene and auto-pick the best on identity + flattering + artifact checks (reject bad hands, gloss, oversized head, drift).
- **Mandatory labelling.** Every output is labelled “AI-generated from verified photos.”

Decision, cost & next steps

Engine: Gemini 2.5 Flash Image, multi-reference, edit-framing + flattering dial, with the guardrail pipeline above.
No per-person training for the MVP.

Cost: ~\$0.04/image. At 3 scenes × a few candidates each that is ~\$0.12–0.36 per man → **~\$120–360 per 1,000 men**, further reduced by generating lazily (only for men who complete onboarding / get viewed). Comfortably within MVP budget.

Tiering: free / MVP users get this zero-shot engine; per-person **fine-tune (Astria)** is offered later as a paid “Pro photos” tier for maximum fidelity.

Next phase (implementation): swap the live /api/photo-enhance from fal.ai → this Gemini-edit recipe, and build the guardrail pipeline (reference selection, per-man vision read, generate-N-pick-best, styling rules). This R&D is complete; the build is separate.